SHANILA JACOBO MALESA

Computer Science Student · University of Birmingham
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PERSONAL STATEMENT

Second-year Computer Science student at the University of Birmingham with a strong practical foundation in software development, built through hands-on projects alongside academic study. Experienced in designing and building projects that integrate the WhatsApp Cloud API, M-Pesa payments, PostgreSQL, USSD and React. Passionate about building technology that solves meaningful problems, with a particular interest in fintech, developer tooling and scalable backend systems. Seeking an internship or placement where I can contribute immediately while continuing to grow.

EDUCATION

BSc Computer Science 2025 – 2028 (expected)
University of Birmingham, Birmingham, UK

Foundation Certificate in Science & Engineering — 78% 2025
Kaplan International College, London, UK

High School Diploma — Grade B 2024
The Great St Albert Girls High School, Kenya

TECHNICAL SKILLS

Languages	JavaScript (Node.js), SQL
Frameworks	Express, React, Next.js (App Router, Server Actions), Tailwind CSS, CSS
Databases	PostgreSQL (schema design, transactions, row-level locking, Row-Level Security), Redis
Backend	REST APIs, layered architecture (routes → services → repositories), webhooks, USSD/chat state machines, PDFKit
APIs	M-Pesa Daraja (STK Push, callbacks), WhatsApp Cloud API, Telegram Bot API, Africa's Talking (USSD/SMS)
Security	JWT, bcrypt, HMAC webhook verification
Tools	Git/GitHub, ngrok, Docker (learning)

PROJECTS

EduLevy — Multi-Tenant School Fee Platform 2026 · in progress
Next.js · Express · PostgreSQL · Redis · M-Pesa Daraja | github.com/shanilamalesa/Edulevy-project

- Designed a multi-tenant PostgreSQL schema where each school's data is separated with Row-Level Security policies, so the database itself stops one school from reading another's records.
- Built the authentication routes and fee/charge REST endpoints using a layered routes → services → repositories structure.
- Integrated M-Pesa STK Push so parents can pay fees their phone, with verified webhooks and duplicates-safe payments records.

WhatsApp CRM + M-Pesa Payments Platform 2025
Node.js · PostgreSQL · WhatsApp Cloud API · M-Pesa Daraja · PDFKit · React | https://github.com/shanilamalesa/WhatsApp-Mpesa-CRM

- Built a CRM that receives WhatsApp messages through webhooks (checked with HMAC signatures), sends automated bot replies, and takes payments with M-Pesa STK Push.
- Handled M-Pesa payment callbacks, with status polling as a fallback, and wrote the payment step so a repeated callback is not recorded twice.
- Generated PDF receipts with PDFKit and sent them to customers on WhatsApp once a payment was confirmed.
- Organised the backend in layers (routes → services → repositories) with JWT login and role-based access, plus a React dashboard showing payments and conversations.

Boda Dispatch — WhatsApp Delivery CRM 2025
Node.js · PostgreSQL · WhatsApp Cloud API · JWT · React | https://github.com/shanilamalesa/A-WhatsApp-Delivery-Request-CRM

- Integrated the dispatch system directly into WhatsApp using the WhatsApp Cloud API.
- Moved the app from SQLite to PostgreSQL and used transactions with row-level locking, so two riders cannot claim the same delivery at the same time.
- Wrote a round-robin algorithm that shares deliveries between riders, with separate access for dispatchers and riders.

ACTIVITIES & INTERESTS

- Demonstrated a WhatsApp Cloud API lead-capture bot at a corporate tech event.
- Canva GenC Student Hackathon 2025, University of Birmingham: in a team of four, analysed a nonprofit's branding problem, built a full pitch deck with Canva's AI tools and presented our solution to the judges, all in one day.

Interests: Fintech and backend systems · Open source · Running and fitness · African tech ecosystems